

Input Absence–Output Inaccessibility Model

***Lifecycle Exclusion, Visible-Field Adequacy, and the
Reproduction of Invisibility***

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Abstract

This document formalizes the **Input Absence–Output Inaccessibility Model** as a structural framework for identifying double exclusion across a system lifecycle. The model applies when the same subjects, conditions, cases, languages, or positions are weakly represented or absent from the inputs that shape a system and are also unable to access, understand, contest, or benefit from the outputs the system produces.

The system may appear effective because it performs well for subjects already visible to its data, categories, records, maps, registries, or assumptions and reachable through its delivery channels. Excluded subjects can remain outside both the performance field and the evaluation field. Their absence is not counted as failure because the system measures only what it can already see and reach.

The model distinguishes data distortion from input absence, output error from output inaccessibility, and system performance from field coverage. It identifies visible-field adequacy, input-channel reproduction, output-channel exclusion, contestability failure, institutional reliance lock-in, and false technical adequacy.

The framework is relevant to automated decisions, public services, warnings, eligibility systems, identity infrastructure, language access, health and social data, and other systems whose inputs and outputs depend on pre-existing visibility and access conditions. It does not reduce the mechanism to algorithmic bias or a digital divide. Its focus is the coupled loop through which prior invisibility is reproduced as apparent functionality.

Keywords

Input Absence; Output Inaccessibility; Double Exclusion; Lifecycle Exclusion; Visible-Field Adequacy; Field Coverage; Representation Failure; Access Failure; Contestability; Institutional Reliance; Technical Adequacy; Public Systems; Boundary Failure; Meta-Writing Ecology.

1. Introduction

A system can work well for the field it recognizes and still fail the field it excludes. Conventional performance measures may not reveal the problem because the excluded subjects do not enter the sample, the baseline, the feedback channel, or the complaint process.

The lifecycle structure matters. Absence at input affects what the system can model. Inaccessibility at output affects who can receive benefits, warnings, explanations, or correction opportunities. When the same subjects are affected at both points, the two failures reinforce one another.

This model makes that coupling visible. It asks who enters system formation, who can access system effects, and whether evaluation includes those who are absent from both.

2. Core Definition

Input Absence–Output Inaccessibility Model describes a structural condition in which the same subjects, cases, positions, or conditions are excluded at two connected points of a system lifecycle:

input formation
and
output access

The model identifies a double exclusion pattern.

A subject may be absent from the data, categories, assumptions, records, maps, registries, or baselines that shape a system. The same subject may then also be unable to access, receive, understand, contest, or benefit from the outputs produced by that system.

The system may still appear functional because it performs adequately for those already visible to its input layer and reachable by its output channels.

The failure is not only bias, error, or poor delivery.

It is a structural loop in which exclusion is reproduced as technical adequacy.

3. Classification and Scope

This document presents **Input Absence–Output Inaccessibility Model** as an external-facing conceptual model. A model in this context identifies a repeatable structural mechanism, its components, sequence, failure patterns, and limits. It is not an empirical law, a predictive score, a software specification, or a compliance standard.

The model is intended to support description, comparison, and diagnostic inquiry across domains. Application to a specific case requires domain evidence and should not be inferred from conceptual resemblance alone. The public version omits non-public editorial records, private source materials, non-public working materials, and implementation details.

4. Central Distinctions

This model distinguishes:

data bias
from
input absence

Data bias distorts representation.

Input absence means the subject is not meaningfully represented at all.

It also distinguishes:

output error
from
output inaccessibility

An output may be technically correct within the system's visible field but inaccessible to the subjects most affected by the system.

It further distinguishes:

system performance
from
field coverage

A system may perform well for the reachable field while failing the excluded field.

5. Core Mechanism and Structural Sequence

5.1 Core Mechanism

The model operates through **lifecycle exclusion coupling**.

A system is built or operated from available input.

Available input reflects prior visibility conditions.

The system then generates outputs through channels that also depend on existing access conditions.

If the same subjects are missing from the input layer and unreachable through the output layer, the system completes a loop:

- prior invisibility
 - input absence
 - partial system model
 - output generation
 - inaccessible delivery
 - institutional reliance
 - exclusion stabilization

The system appears to work because it works inside the already-visible field.

5.2 Structural Sequence

1. A field contains subjects, cases, needs, risks, languages, positions, or conditions.
 2. Some subjects are already weakly recorded, unrecorded, miscategorized, or structurally absent.
 3. A system is trained, calibrated, designed, or operated using available input.
 4. Available input reproduces prior visibility boundaries.
 5. The system generates outputs such as decisions, warnings, classifications, recommendations, aid routing, eligibility signals, risk scores, summaries, or priority lists.
 6. These outputs are delivered through existing access channels.
 7. The same subjects missing from input may also lack access to the output channel.
 8. The system output becomes actionable for visible and reachable subjects.
 9. Excluded subjects remain outside recognition and outside delivery.
 10. The system's partial coverage is mistaken for functional adequacy.
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6. Structural Components

The structure should be examined through the object being acted on, the roles involved, the relevant authority boundary, the evidence available at each node, the direction of consequence, the possibility of correction, and the conditions under which the relation becomes stable.

7. Primary Failure Patterns

1. Visible-Field Adequacy

- system works for visible subjects
 - performance appears strong
 - invisible subjects are not measured as failure

The system's success metrics are anchored to the already-covered field.

2. Input-Channel Reproduction

- available data reflects prior exclusion
- model treats available data as field reality
- absent subjects remain structurally unmodeled

The system inherits earlier absence as if it were neutral input.

3. Output-Channel Exclusion

- output is generated
- delivery requires access to existing channels
- excluded subjects cannot receive or use the output

The system produces results that cannot reach the subjects most affected.

4. Contestability Failure

- decision affects subject
- subject lacks channel to understand, challenge, or correct it
- system result becomes uncontestable

The failure is not only absence from benefit, but absence from correction.

5. Institutional Reliance Lock-In

- institution relies on system output
- output excludes missing subjects
- exclusion becomes administrative reality

The institution treats the partial system as a complete decision surface.

6. False Technical Adequacy

- system metrics appear acceptable
- excluded population is outside measurement
- failure is not visible to evaluation

The system is evaluated only where it can already see.

8. Operational Evaluation

1. Who or what is absent from the input field?
2. How was the relevant field defined?
3. Which outputs affect the absent subjects?
4. Can those subjects receive, understand, use, or contest the outputs?
5. Are performance metrics calculated only on visible and reachable cases?

6. Do compensating channels represent or reach the excluded field?
 7. Does institutional reliance turn partial coverage into administrative reality?
 8. How can absence be measured without requiring prior system visibility?
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9. Application Domains

1. Automated eligibility and allocation

People missing from records or categories may also lack access to notices, appeals, or benefits.

2. Public warning systems

Populations absent from risk maps or communication planning may not receive the warnings generated from those systems.

3. Language and accessibility infrastructure

Unsupported languages or access needs can be missing from input design and from output delivery.

4. Health and social data

Under-recorded groups may be poorly modeled and less able to receive or contest resulting classifications.

5. Platform and identity systems

Subjects without recognized identifiers may be excluded from both system representation and service access.

10. Relations to Adjacent Public Frameworks

1. False Legibility Model

False Legibility Model describes how readable surfaces can be accepted before their underlying structure is understood. The present entry identifies a concrete way in which fluency, familiarity, or interface clarity can conceal an unresolved boundary.

2. Cost Visibility and Redistribution Model

Cost Visibility and Redistribution Model tracks how costs become visible at one node and hidden or displaced at another. The present entry shows how verification, correction, access, or responsibility costs can be redistributed by the relevant interface.

3. Boundary Failure Diagnostics

Boundary Failure Diagnostics provides a general vocabulary for locating non-entry, closure, lock-in, and boundary loss. The present entry narrows that diagnostic field to a repeatable mechanism.

4. Responsibility Alignment Model

Responsibility Alignment Model clarifies whether ownership, authority, capacity, visibility, and accountability remain attached to compatible nodes. The present entry specifies one condition under which that alignment can fail or become nominal.

5. Observer-Representation Boundary

Observer-Representation Boundary separates what an observer can read from what a representation can legitimately establish. The present entry specifies the evidential or operational consequences of crossing that boundary.

11. Boundary Conditions and Non-Applicability

11.1 Boundary Conditions

This model applies when:

1. A system depends on data, records, categories, registries, maps, identity layers, histories, or assumptions.
2. Some subjects or conditions are absent or weakly represented in that input layer.
3. The system produces outputs that affect access, recognition, allocation, warning, routing, eligibility, or decision-making.
4. The output channel is itself access-dependent.
5. Subjects missing from input are also unable to receive, understand, contest, or benefit from outputs.
6. Institutional or platform reliance treats the system as adequate despite field exclusion.

This model is weak or absent when:

1. Input coverage is structurally complete for the relevant field.
2. Output access is universal or independently verified.
3. The system has no downstream consequence for excluded subjects.
4. Missing subjects are explicitly represented through compensating mechanisms.
5. The output can be contested through accessible channels.
6. Evaluation includes the excluded field rather than only the visible field.

11.2 Non-Applicability

This model should not be reduced to:

algorithmic bias
digital divide
data quality problem
software performance issue
interface design failure
accessibility compliance
statistical underrepresentation
communication failure
policy failure

These may be application surfaces.

They are not the source ontology of the model.

The model concerns a structural lifecycle loop in which absence at the input layer and inaccessibility at the output layer reinforce one another.

12. Public Orientation and Method Boundary

This document is designed as a public conceptual entry. It provides a definition, mechanism, diagnostic orientation, and limit statement. It omits non-public editorial records, private source materials, non-public working materials, and implementation details.

The entry is not a substitute for empirical research, legal analysis, professional judgment, technical testing, or domain-specific governance. It should not be used to score people or institutions, automate high-stakes decisions, or infer misconduct from conceptual similarity.

Examples and application domains are illustrative. A case should be classified under **Input Absence–Output Inaccessibility Model** only when the relevant structure can be demonstrated from available evidence.

13. Limitations

1. Not every absence is exclusion

A subject may be outside the intended scope of a system for legitimate reasons.

2. Coverage can be difficult to define

The relevant field must be specified rather than assumed universal.

3. Compensating mechanisms vary

Representation and access can be supported through different channels.

4. No fairness metric

The model does not provide statistical parity or legal discrimination tests.

5. Output access is multidimensional

Delivery, comprehension, usability, contestability, and benefit are distinct.

14. Minimal Formulation

A system cannot serve what it does not see,
and cannot protect what it cannot reach.

Compressed form:

Input absence plus output inaccessibility produces exclusion through
apparent functionality.

15. Conclusion

Input absence and output inaccessibility become a coupled failure when the
same subjects are missing from system formation and unreachable in system
operation. The system then stabilizes exclusion through metrics that describe only
the visible field.

A structurally adequate assessment compares performance with field coverage,
tests independent access and contestability, and treats absent subjects as part of the
evaluation problem rather than outside it.

Naming Declaration

Input Absence–Output Inaccessibility Model is designated within Meta-
Writing Ecology as a structural model for identifying double exclusion across system
lifecycle stages.

The name refers to the coupling between missing input representation and
inaccessible output delivery.

It does not refer to any specific population, institution, platform, technology, policy domain, dataset, report, or real-world case.

The model should be used when the central issue is not only that a system contains incomplete data, and not only that its outputs are difficult to access, but that the same absence is reproduced across both sides of the system:

not represented in formation
+
not reachable in operation
=
exclusion stabilized as functionality

The term preserves the distinction between representation failure, access failure, and the structural loop formed by their combination.